

Sports & Entertainment: Match Outcome Prediction Using Logistic Regression and Random Forest

PROBLEM STATEMENT

Predicting the outcome of an Indian Premier League (IPL) cricket match is framed in this project as a supervised binary classification problem. Given a set of pre-match and historical performance features for two competing teams — including recent team form, head-to-head record, venue and home/away advantage, and toss outcome — the objective is to predict whether a specified team will win or lose the match. The target variable is the match result (win = 1, loss = 0) for the chosen team, as recorded in the winner column of the dataset.

This problem is significant because match outcomes in cricket are influenced by a large number of interacting factors that are difficult to weigh manually — such as toss decisions, venue-specific batting/bowling advantages, and a team's current form — and traditional prediction methods relying on commentator intuition or simple statistics (e.g., win percentage) fail to capture these interactions. A data-driven machine learning approach can uncover such patterns systematically, offering practical value to fans, broadcasters, team management, and sports analytics platforms. This project further aims to compare an interpretable baseline model (Logistic Regression) against a more powerful ensemble model (Random Forest) to evaluate the trade-off between interpretability and predictive accuracy for this specific domain.

DATASET IDENTIFICATION

The project uses the IPL Complete Dataset (2008–2024), publicly available on Kaggle and authored by Prateek Bhardwaj. The dataset covers every IPL season from the tournament's inception in 2008 through 2024, comprising over 1,000 matches and more than 260,000 individual deliveries, and is provided as two linked CSV files:

File	Description	Key Columns
matches.csv	Match-level summary for every IPL match played, 2008–2024	season, city, venue, team1, team2, toss_winner, toss_decision, winner, result margin, player of match
deliveries.csv	Ball-by-ball record for every delivery bowled across all matches	match_id, inning, batting_team, bowling_team, over, ball, batter, bowler, total_runs, is_wicket, dismissal_kind

The two files are linked via a common `match_id`, allowing match-level outcomes in `matches.csv` to be enriched with granular, ball-by-ball statistics aggregated from `deliveries.csv` (e.g., team run rate, powerplay performance, wicket-loss patterns). This combination makes the dataset well suited to engineering features such as recent-form indicators, head-to-head win ratios, and venue-based performance trends, which are used as inputs to the classification models.

Suitability of the Dataset

- Covers the complete IPL history (2008–2024), giving sufficient volume and variety of matches, teams, and venues for robust model training.
- Provides both match-level (`matches.csv`) and micro-level (`deliveries.csv`) data, enabling rich, engineered features rather than relying on raw columns alone.
- Clearly identifiable target variable (match winner) suitable for binary classification per team.
- Publicly available and well documented on Kaggle, ensuring reproducibility.

Known Limitations

- Team names and franchises have changed over seasons (e.g., relocations, rebranding), requiring careful standardisation during cleaning.
- Rare match outcomes such as ties, no-results, or matches decided by the Duckworth–Lewis–Stern (DLS) method need separate handling during pre-processing.
- Possible class imbalance and home-ground bias, which will be examined during exploratory data analysis (EDA) .